MYSQL 8.0+ CAPSTONE PROJECT • STUDENT ASSIGNMENT BRIEF

RetailPulse — E-Commerce Sales, Cohort & RFM Analytics

End-to-End Relational Data Modeling, Data Quality Auditing, KPI Analysis, Customer Segmentation & Production BI Layer

ROLE SIMULATED	DATABASE ENGINE	DATA PERIOD	TOTAL QUESTIONS
Data Analyst	**MySQL 8.0+**	**Jan 2024 – Oct 2026**	**5 Audit + 24 SQL + 8 BI**

1. Business Context & Objectives

You have joined **RetailPulse**, a multi-category omnichannel e-commerce company operating across five international regions (North America, Europe, South Asia, Southeast Asia, and Middle East). Leadership wants a comprehensive SQL analytics layer to evaluate revenue quality, category profitability, delivery SLA performance, marketing campaign ROI, customer retention, RFM value segments, and inventory risk.

> **Your Mission:** Build the `retailpulse_dw` database from the provided CSV files, audit the data for anomalies, answer all **24 analytical business questions** using pure MySQL 8.0 SQL, create reusable BI Views/Procedures/Triggers, and summarize actionable business recommendations.

2. Provided Files & Dataset Dictionary (Import Order)

You are provided with `00_student_schema.sql`, `01_load_csv_data.sql`, and **9 raw CSV files** inside `csv_data/`. Because foreign keys enforce referential integrity, import the CSV files strictly in the parent-to-child order below:

Order	CSV File / Table	Rows	Primary / Foreign Keys	Business Description
1	`regions.csv`	5	PK: `region_id`	Operating territories, country groups, and regional heads.
2	`customers.csv`	30	PK: `customer_id`; FK: `region_id`	Customer demographics, signup date, acquisition channel, loyalty tier.
3	`categories.csv`	6	PK: `category_id`	6 product categories with department and target gross margin %.
4	`suppliers.csv`	6	PK: `supplier_id`	Vendors with country, lead time (days), and reliability score.
5	`products.csv`	24	PK: `product_id`; FK: `category_id`, `supplier_id`	SKU catalog, unit cost, unit price, stock on hand, reorder level.
6	`campaigns.csv`	8	PK: `campaign_id`	Marketing campaigns, channel, active dates, spend budget, promo code.
7	`orders.csv`	72	PK: `order_id`; FK: `customer_id`, `campaign_id`, `region_id`	Order header with order/ship/delivery timestamps, status, payment, order discount.
8	`order_items.csv`	159	PK: `order_item_id`; FK: `order_id`, `product_id`	Line-item granularity: quantity, unit price, unit cost, line discount.
9	`returns.csv`	10	PK: `return_id`; FK: `order_id`, `order_item_id`	Returned items, return date, reason, refund amount, restock flag.

Setup Instructions (Choose One Import Method)

1. Open **MySQL Workbench 8.0** (or MySQL CLI) and execute `00_student_schema.sql` to create `retailpulse_dw`.
2. **Option A (Fast SQL Loader):** Open `01_load_csv_data.sql`, update the folder path if needed (keep forward slashes `/`), and execute with `local_infile` enabled.

3. **Option B (Workbench GUI):** Right-click each table in `retailpulse_dw` → *Table Data Import Wizard* → select the matching CSV in order 1–9.

> **Important Financial Definitions for All Queries:**
> - Exclude `order_status = 'Cancelled'` from sales, revenue, margin, and basket calculations unless specifically analyzing cancellations.
> - **Gross Merchandise Value (GMV):** `SUM(quantity * unit_price)`.
> - **Net Sales Before Refunds:** `SUM(quantity * unit_price - line_discount) - order_discount`.
> - **Realized Net Revenue:** `Net Sales Before Refunds - SUM(refund_amount)`.
> - **Gross Profit & Margin %:** `Realized Net Revenue - COGS`; Margin % = `100.0 * Gross Profit / Realized Net Revenue`.
> - **Avoid Join Fan-Out:** Never join `orders`, `order_items`, and `returns` directly in the same aggregation without pre-aggregating line-level tables in CTEs!

3. Part A — Data Cleaning & Integrity Audit (5 Tasks)

Save these queries in `01_data_audit.sql`. Before analyzing business KPIs, audit the database to prove data completeness and financial consistency:

Task A1: Table Row Count Inventory `UNION ALL`

Write a single query returning the table name and row count across all 9 populated tables (`regions` through `returns`).

Task A2: Missing & Invalid Critical Fields Audit `CASE Aggregation`

Check `customers` for blank names, malformed emails (not matching `%@%.%`), or missing birth/signup dates. Check `orders` for `Delivered` orders missing `delivered_date`, `Cancelled` orders with a non-null `shipped_date`, or negative discounts/shipping fees.

Task A3: Duplicate Detection `GROUP BY / HAVING`

Verify there are zero duplicate customer `email` values, zero duplicate product `sku` codes, and check whether any single order repeats the same `product_id` across multiple `order_items` lines.

Task A4: Chronological & Over-Refund Sanity Checks `Multi-Table Joins`

Detect: (1) orders placed before a customer's `signup_date`, (2) orphan orders with no line items, (3) returns where `return_date < DATE(delivered_date)` or `order_id` mismatches, and (4) refunds exceeding either the line item net value or total order net paid after `order_discount`.

Task A5: Consolidated Audit Health Scorecard `PASS / REVIEW`

Combine your critical validation checks into a summary table outputting `audit_check`, `issue_count`, and `status` (`'PASS'` if 0, else `'REVIEW'`).

4. Part B — Core Business & Executive KPI Analysis (Questions Q1–Q12)

Save these queries in `02_kpi_analysis.sql`. Clearly label each query with a comment header.

Q1. Executive Summary KPI Scorecard `CTEs • Aggregate KPIs`

Produce a one-row executive summary showing: Total Orders, Active Customers, Delivered Orders, Cancelled Orders, Fully Returned Orders, Cancellation Rate %, GMV, Total Refunds, Realized Net Revenue, Total COGS, Gross Profit, Gross Margin %, and Average Order Value (AOV based on net sales before refunds).

Q2. Monthly Revenue Trend, Profit & MoM Growth % `LAG() • DATE_FORMAT`

Calculate monthly order count, net revenue (net of discounts), gross profit, prior month's revenue, and Month-over-Month (MoM) revenue growth % ordered chronologically from `2024-01` to `2026-10`.

Q3. Year-over-Year (YoY) Quarterly Performance Comparison `PARTITION BY Quarter`

Aggregate orders, quarterly revenue, quarterly profit, and margin % by Year and Quarter (`Q1` – `Q4`). Compare each quarter against the same quarter of the prior year using `LAG()` to compute YoY Revenue Growth %.

Q4. Category Performance vs. Target Margin & Share % `SUM() OVER ()`

For each product category, calculate units sold, distinct orders, gross revenue, refunds, net revenue, gross profit, actual margin %, variance vs. `target_margin_pct` (in percentage points), and share of total company net revenue %.

Q5. Regional Revenue & Delivery SLA Performance `TIMESTAMPDIFF • DATEDIFF`

Rank regions by net revenue and compute: total orders, non-cancelled orders, average days to ship (in decimal days via hours/24), average days to deliver, count of late deliveries (`DATEDIFF(delivered_date, order_date) > promised_days`), and Late Delivery %.

Q6. Payment Method Mix, AOV & Cancellation / Return Rates `Window Share %`

Analyze each `payment_method` : order count, % share of total orders, net revenue, AOV, cancellation rate %, and full-order return rate %. Identify which payment method has the highest failure/return risk.

Q7. Top 10 Products by Net Revenue vs. Profit Rank `DENSE_RANK()`

List the top 10 products by net revenue (net of line discounts and refunds), showing SKU, category, units sold, net revenue, gross profit, profit margin %, overall Revenue Rank, and overall Profit Rank. Highlight any high-revenue, low-margin anomalies.

Q8. Supplier Scorecard & Quality Risk `Multi-Table Join`

Evaluate all 6 suppliers by active SKUs, units sold, gross sales, returned item count, and item return rate %, alongside their contracted `lead_time_days` and `reliability_score` .

Q9. Discount Impact Analysis (Discounted vs. Full Price) `Conditional Bucketing`

Compare `'Discounted'` orders (any order or line discount > 0) against `'Full Price'` orders across: order count, average gross basket, average discount given, average net order value, total net revenue, gross profit, and gross margin %.

Q10. Marketing Campaign ROI & ROAS `Campaign Attribution`

For all 8 marketing campaigns (including any with zero orders if applicable via `LEFT JOIN`), compute campaign spend, attributed orders, unique buyers, attributed realized net revenue, gross profit generated, **ROAS** (`Net Revenue / Budget`), and **Net Marketing ROI %** (`100 * (Gross Profit - Budget) / Budget`).

Q11. Return Reason Breakdown & Financial Leakage `Share % • Write-Offs`

Group `returns` by `return_reason` to show return count, % share of returns, total refund leakage, restocked item count, and damaged/non-restocked write-offs.

Q12. Day-of-Week Ordering & Revenue Patterns `DAYOFWEEK • DAYNAME`

Aggregate non-cancelled orders by day of week (Sunday through Saturday in proper calendar order) showing total orders, total revenue, and average order value.

5. Part C — Advanced SQL: RFM, Cohorts & Market Basket (Questions Q13–Q24)

Save these queries in `03_advanced_analytics.sql`. Use CTEs and MySQL 8.0 Window Functions throughout.

Q13. Customer Lifetime Value (CLV) Leaderboard & Regional Rank `DENSE_RANK()`

Rank the top 15 customers by Realized LTV (net sales minus refunds across all non-cancelled orders). Display customer details, order count, first/last order dates, realized LTV, AOV, global LTV rank, and regional LTV rank.

Q14. Full RFM Segmentation Using `NTILE(5)` `NTILE(5) • CASE Segmentation`

Using the maximum non-cancelled order date in the dataset as the reference `as_of_date`, compute for every active buyer: **Recency (days)**, **Frequency (distinct orders)**, and **Monetary (realized net spend)**. Score each from 1 to 5 via `NTILE(5)` (5 = most recent, most frequent, highest spend) and classify into segments:

`Champions` (R≥4, F≥4, M≥4) • `Loyal Customers` (R≥3, F≥3, M≥3) • `Recent / Potential Loyalists` (R≥4, F≤2) • `At Risk` (R≤2, F≥3) • `About to Sleep` (R=2, F≤2) • `Hibernating / Lost` (R=1, F≤2) • `Needs Attention` (all other combinations).

Q15. RFM Segment Revenue Contribution Summary `Nested Aggregation`

Roll up your Q14 RFM output by `rfm_segment` to display: customer count, % of active customer base, average recency, average frequency, average LTV, segment total revenue, and % of total company revenue.

Q16. Quarterly Cohort Retention Matrix `TIMESTAMPDIFF(QUARTER)`

Assign each buyer to a cohort based on the calendar quarter of their **first non-cancelled order** (`2024-Q1` through `2026-Q3`). Pivot retention rates (% of cohort active) across `Q+1` , `Q+2` , `Q+3` , `Q+4` , and `Q+5+` quarters.

Q17. Repeat Purchase Gap Analysis `LAG() • DATEDIFF()`

For customers with 2+ orders, use `LAG(DATE(order_date))` to compute the days elapsed between consecutive orders and summarize each customer's repeat transitions, minimum gap, average gap, and maximum gap.

Q18. Personalized Churn Risk Scoring `Overdue Ratio`

Compare each repeat customer's `days_inactive` (since last order relative to dataset max date) against their own personal `avg_cycle_days` . Flag: `High Churn Risk` (> 2.0× cycle), `Watchlist` (> 1.25× cycle), or `Healthy Cadence` .

Q19. ABC Product Classification (Pareto 80/20 Analysis) `Running Sum Window`

Compute each product's net revenue, % share, and cumulative revenue % ordered descending. Classify into `Class A (Core Stars)` (≤ 70% cumulative), `Class B (Moderate)` (≤ 90%), and `Class C (Long Tail)` (> 90%).

Q20. Market Basket Analysis (Product Affinity Pairs) `Self-Join • Support / Confidence / Lift`

Find the top 12 pairs of distinct products bought together in at least 2 non-cancelled orders. Calculate: `co_purchase_count` , `support_pct` , `confidence_a_to_b_pct` , `confidence_b_to_a_pct` , and `lift_score` .

Q21. Cumulative Revenue & 3-Month Rolling Moving Average `ROWS BETWEEN 2 PRECEDING`

Compute monthly revenue alongside running cumulative revenue from `2024-01` onward and a 3-month rolling average using window frame clauses.

Q22. Acquisition Channel Quality & Repeat Conversion `LEFT JOIN Customers`

Evaluate all 6 customer `acquisition_channel` values across: total acquired customers (including those with 0 orders), repeat buyers (2+ orders), repeat buyer rate %, average orders per customer, average LTV, and total channel LTV.

Q23. Inventory Stock-Out Risk & Supplier Reorder Alert `180-Day Velocity`

Compute trailing 180-day sales velocity (relative to dataset max order date) and estimated days of stock cover per SKU. Flag SKUs as `REORDER NOW` (stock ≤ reorder level), `LOW BUFFER` (days of cover ≤ 2× supplier lead time), or `HEALTHY`.

Q24. Customer Order Conversion Funnel `Funnel Drop-Off`

Across all 30 signed-up customers, measure how many placed ≥ 1 order, ≥ 2 orders, and ≥ 3 orders, along with step-to-step conversion percentages.

6. Part D — Production BI Layer (Views, Function, Procedures & Trigger)

Save these database objects in `04_bi_objects.sql` so BI tools (Power BI / Tableau / Excel) can consume standardized metrics:

Tasks B1–B8: Reusable MySQL Objects `Production SQL`

- **B1 (Stored Function):** `fn_profit_margin_pct(p_revenue, p_cost)` — deterministic, zero-division-safe margin % calculator.
- **B2 (View):** `vw_order_financials` — order-level GMV, total discounts, refunds, realized net revenue, COGS, profit, margin %, actual delivery days, and SLA status (`On Time`, `Late`, `In Transit / Processing`, `Cancelled`).
- **B3 (View):** `vw_customer_360` — full customer summary with RFM scores (partitioning active buyers separately from zero-order signups) and RFM segment label.
- **B4 (View):** `vw_product_performance` — SKU-level sales, net revenue, refunds, COGS, stock levels, and item return rate %.
- **B5–B7 (Stored Procedures):** `sp_get_executive_kpis(start_date, end_date)`, `sp_get_category_drilldown(category_name, year_or_null)`, and `sp_get_customers_by_rfm_segment(segment_name)`.
- **B8 (Trigger):** `trg_after_order_item_insert` — decrements `products.stock_qty` safely (non-negative) and logs a row in `inventory_audit_log` when a non-cancelled order item is inserted.

7. Part E — Executive Insights & Grading Rubric (100 Marks)

Submit a 1-page insights brief (PDF or Markdown) answering: (1) Which category misses its target margin and why? (2) Which region & payment method suffer the highest SLA/cancellation risk? (3) Which customer segments drive >50% of revenue and who needs win-back outreach? (4) Which product bundles should be promoted from Market Basket Lift? (5) Which 4 SKUs require immediate restocking?

Evaluation Component	Weight	Grading Criteria
Database Setup & Audit (Part A)	15 Marks	Clean schema creation, full CSV import (all 9 tables), zero unhandled errors, complete A1–A5 validation checks.
Core KPI Analysis (Q1–Q12)	30 Marks	Accurate revenue/profit logic (excluding cancellations, zero join fan-out), clean window shares & MoM/YoY calculations.
Advanced Analytics (Q13–Q24)	35 Marks	Correct `NTILE(5)` RFM logic, cohort pivoting, `LAG()` cadence churn, Pareto ABC, and Support/Confidence/Lift math.
Production BI Layer (B1–B8)	10 Marks	Working Views, Stored Function, parameterized Stored Procedures, and Trigger tested cleanly with sample calls.
Insights & Presentation (Part E)	10 Marks	Clear business storytelling backed by exact query metrics, ER diagram export from Workbench, and actionable recommendations.